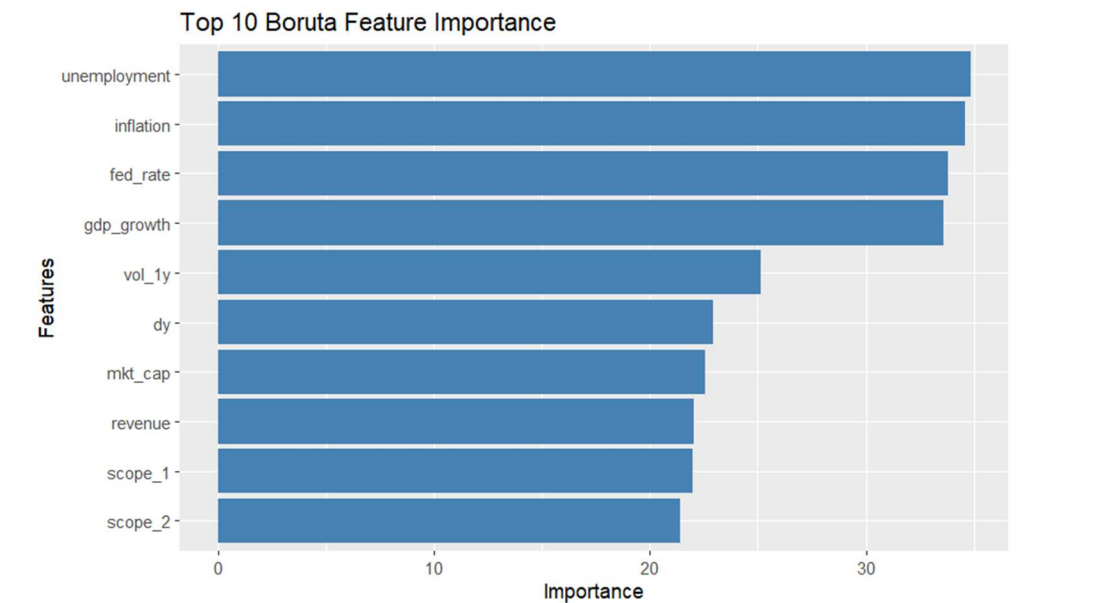


Dynamic Portfolio Backtesting Project

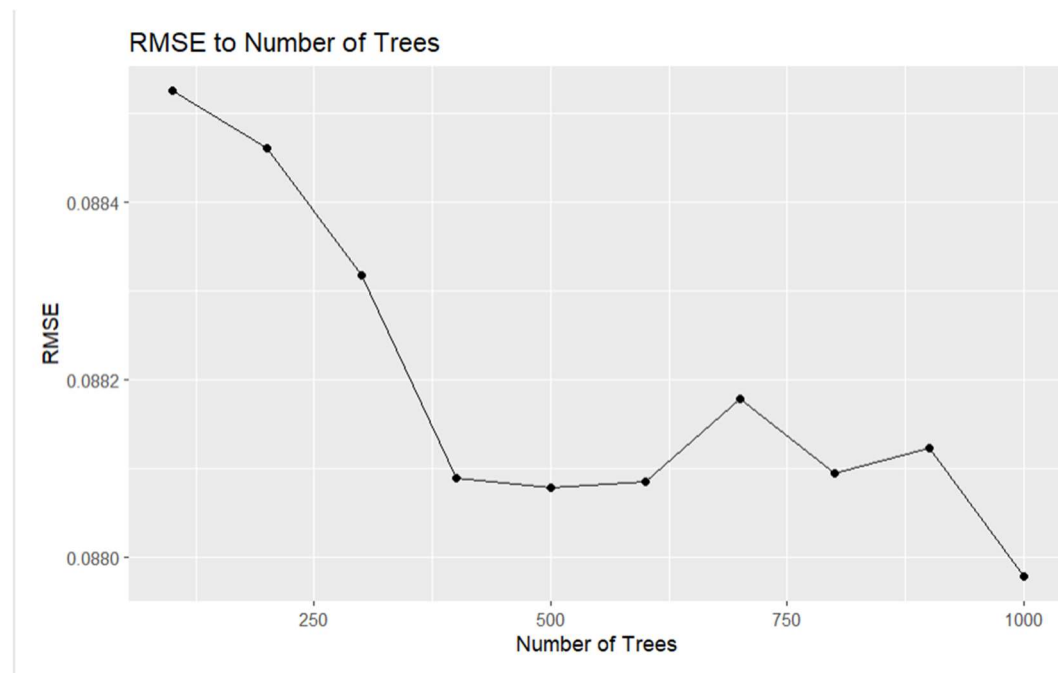
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This project focuses on developing and backtesting dynamic portfolio strategies using machine learning techniques. The main goal is to build a dynamically managed portfolio based on the top 10 stocks in the previous period. Use machine learning to select a suitable model, and expect the model's good prediction results to bring excellent returns.

The project has the following key steps, including feature selection, model training, and portfolio construction based on predicted returns. Initially, I first cleaned the data and selected and standardized key features such as `vol\_1y`, `revenue`, `mkt\_cap`, `rec\_mean`, `p2b`, `d2e`, `prof\_marg`, `dy`, `esg`, `scope\_1`, `scope\_2`, `scope\_3`, `inflation`, `gdp\_growth`, `unemployment`, and `fed\_rate`. Using the Boruta algorithm, the most important features affecting forward returns were identified, including `vol\_1y`, `revenue`, `mkt\_cap`, `rec\_mean`, `p2b`, `d2e`, `prof\_marg`, `dy`, `esg`, and `scope\_1`.



The second step is model training, which is done by splitting the data into training, validation, and test sets in a ratio of 60%-20%-20%. The random forest model is used because of its robustness and ability to handle a large number of features. The optimal number of trees for the model is determined by evaluating the RMSE of different values. The lowest RMSE indicates the highest model complexity. The final model is trained with this optimal number of trees and evaluated on the test set. From the figure, it can be seen that within the displayed range, 100-1000, the decision tree reaches the best value at 1000.

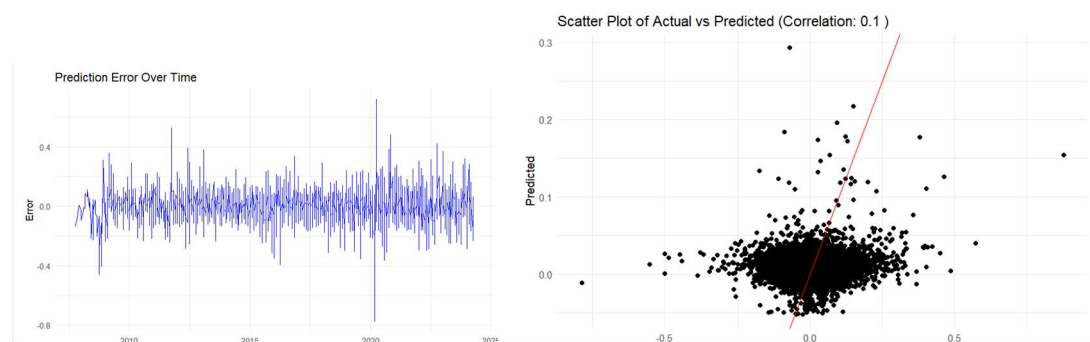


Next, I used several key indicators to evaluate the performance of the model, namely: average return, annual return, annual volatility, Sharpe ratio, maximum drawdown, MAR ratio, turnover rate, and adjusted Sharpe ratio. The results show an average return of 0.0118, an annual return of 0.1418, and an annual volatility of 0.3002. The Sharpe ratio, which measures risk-adjusted returns, is 0.4057, while the maximum drawdown is 0.7326. Another risk-adjusted metric, the MAR ratio, is 0.1662. The portfolio turnover ratio is 7.1057, reflecting the frequency of asset reallocation. The adjusted Sharpe ratio, which takes into account transaction costs, is 0.1217.

average_return<dbl>	annual_return<dbl>	annual_volatility<dbl>	sharpe_ratio<dbl>	max_drawdown<dbl>	mar_ratio<dbl>	turnover<dbl>	adjusted_sharpe_ratio<dbl>
0.01181588	0.1417906	0.3002066	0.4056893	0.7326216	0.1662394	7.105691	0.1216573

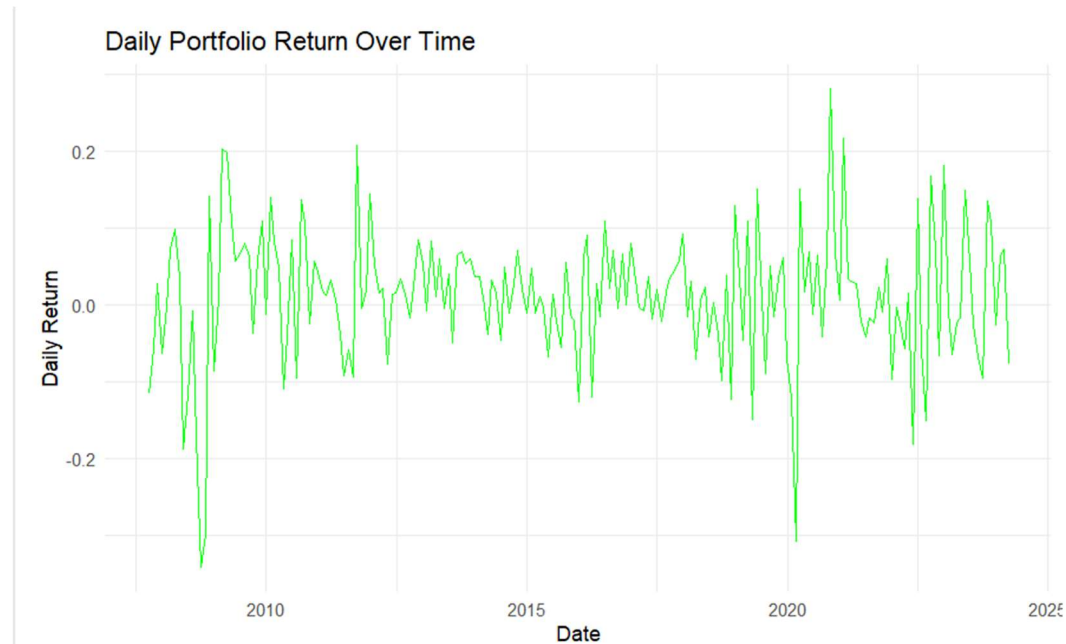
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Visualizing the model predictions versus actual returns over time shows that while the model captures the general trend, there are periods of significant differences, especially during periods of high volatility. The distribution of prediction errors is relatively concentrated around zero, indicating no significant deviations, but there are considerable gaps, indicating some prediction inaccuracies. The scatter plot of actual returns versus predicted returns shows low correlation, highlighting potential areas for improvement in model accuracy.



The dynamic portfolio constructed based on the model predictions shows that the cumulative

returns during the analysis period exceed the initial investment. However, the daily portfolio returns show a lot of variability, indicating that further improvement is needed to stabilize returns.



There are still many areas for improvement, such as the relatively low Sharpe ratio and high maximum drawdown indicate that the risk-adjusted return of the model can be improved. In addition, the high turnover rate means huge transaction costs, which need to be minimized to obtain better net performance. The project can capture complex market dynamics with a sophisticated machine learning model. It will also be beneficial to enhance the model's predictive ability in turbulent market conditions and implement transaction cost optimization strategies.